INTRA-UTERINE RESPIRATION-LIKE MOVEMENTS IN RELATION TO DEVELOPMENT OF THE FETAL VASCULAR SYSTEM

A Discussion of Intra-uterine Physiology Based upon Cases of Congenital Absence of the Trachea, Abnormal Vascular Development, and Other Anomalies *

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Clinical observation, and the experimental work of Snyder and Rosenfeld,[1-3] Barcroft and Barron,[4] and others,[5-7] have established the fact that the fetus makes respiration-like movements in utero. These studies and, more especially, those of Davis and Potter,[8] indicate that such movements are a normal part of fetal activity, and that intra-uterine respiratory movements normally result in the aspiration of amniotic fluid into the fetal lung. Whether or not intra-uterine respiration is a vital function has not been established.

If intra-uterine respiratory activity is necessary for the normal development or survival of the fetus, interference with it should result in abnormal development or death. If it can be shown that interference with intra-uterine pulmonary expansion, and that alone, produces maldevelopment of some organ or system, it will constitute proof that intra-uterine respiratory movements are physiologic. We were unable to find a report of any anomaly that could be shown to result only from interference with intra-uterine pulmonary expansion.

Material

This report is based upon necropsies performed in cases of prenatal and neo-natal deaths in the Augustana Hospital, Chicago. The first 2 cases reported in detail were those of infants who died at birth. Both were victims of anomalies that made respiration impossible. Although the anomalies affecting the respiratory system were dissimilar, the same altered ratio between the third portion of the aortic arch and the ductus arteriosus was present in both infants. The third case was that of a stillborn infant who had a pulmonary anomaly that, in itself, would not prevent respiration but as an anencephalic monster probably lacked central nervous system control for respiratory movement.[9] No vascular anomaly was found in this case.

* Received for publication, February 23, 1949.

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Report of Cases

Case 1

Case 1 was a female infant born on the service of Dr. William Browne. The mother's past history was negative, and the pregnancy, her first, had been without incident. Fetal heart tones were normal throughout labor. Delivery was completed by low forceps. The child appeared normal except that the cardiac impulse was displaced to the right side of the chest. The infant made numerous unsuccessful attempts to breathe. A tracheal catheter was introduced, but even with the catheter in place it was impossible to force air into the lungs.

At autopsy, the body measured 52 cm. from crown to heel. It weighed 3572 gm. and was normally developed and well nourished. The head and upper extremities were cyanotic.

The left diaphragm was absent except for a narrow shelf of muscle which extended onto the anterior surface of the esophagus to which it was attached. Distal to this attachment the esophagus bent sharply upwards and to the left, forming a complete loop. The stomach and spleen were located behind the heart in a hernial sac formed by the mediastinal pleura projecting into the right pleural cavity. The left pleural cavity contained most of the small and large bowel and the left lobe of the liver. The trachea deviated to the right and was patent. The right lung appeared normally developed, but was non-crepitant and firm. The left lung was markedly hypoplastic, and resembled the lung in a similar case reported by Potter and Bohlender.[10]

The heart was displaced to the right, with the apex at about the midsternum. The cardiac chambers were normally formed and situated. The foramen ovale was anatomically patent. The great vessels arose normally. The ascending aorta was 7 mm. in diameter, but after giving off the usual three branches from the arch, the third portion of the arch was only 5 mm. in diameter. The ductus arteriosus was 8 mm. in diameter, 12 mm. in length, and was patent. The descending aorta was 10 mm. in diameter. The ductus arteriosus exceeded the diameter of both the ascending aorta and the third portion of the arch, the latter by a ratio of 8 to 5 ($I/D^* = 0.62$). This child was unable to breathe after birth by reason of the absence of the diaphragm, which made it impossible to produce a negative pressure in the chest cavity. It is reasonable to suppose that intra-uterine respiration also was impossible.

Potter[11] has observed elongation and narrowing of the aortic isthmus in cases of diaphragmatic hernia in neonatal autopsies. No measurements of the vessels were recorded. According to Potter and Bohlender,[10] a fetal lung that fails to develop grossly as a result of external pressure

* Diameter of the isthmus : diameter of the ductus arteriosus.

from herniated abdominal contents or other causes is also retarded in its histologic development. It exhibits alveolar and vascular growth normal to its size, rather than to the chronologic age of the fetus.

The embryogenesis of the diaphragm is an interesting and complicated process, but since it is of incidental interest here, discussion of it will be omitted. An excellent article on this subject is presented by Bremer.[12]

Case 2

Case 2 was a male infant born at term on the service of Dr. Crean. The mother's past history was negative, and this, her first pregnancy, had been without incident. Fetal heart tones were of good quality and remained so throughout the uneventful labor. Delivery was spontaneous. The infant's heart rate and rhythm were normal, but he made only occasional attempts to breathe, and it was noted that the chest failed to expand with respiratory efforts. Repeated attempts to pass a tracheal catheter were unsuccessful. The child became progressively more cyanotic and died in 20 minutes.

At autopsy, the body was that of an apparently well developed white male infant, 48 cm. in length and weighing 3226 gm. The skin was pale, and no external abnormality was found except bilateral cryptorchidism.

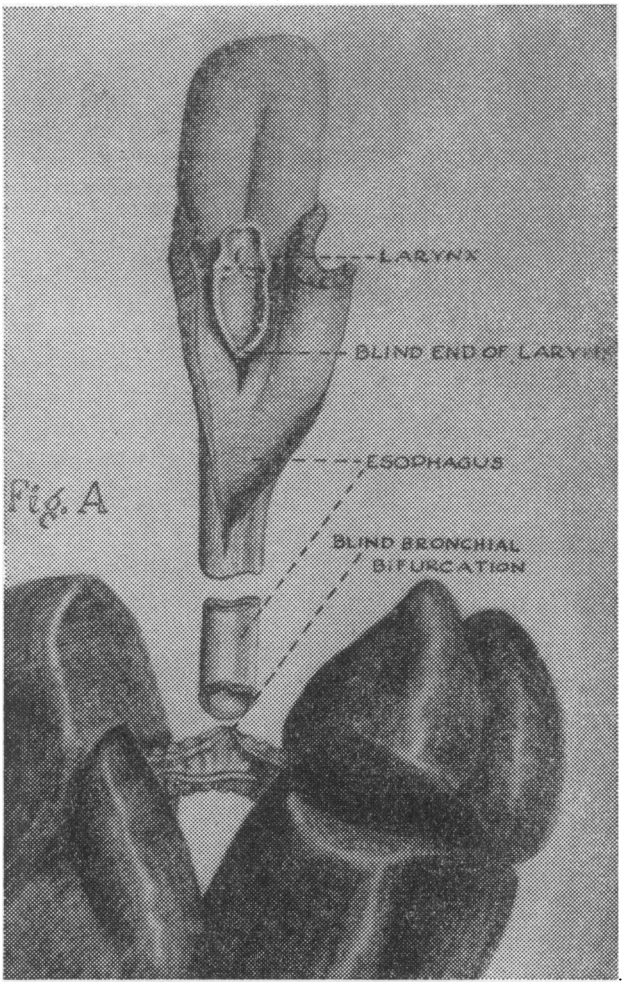
The chest cavity contained a small amount of straw-colored fluid. The lungs were atelectatic and dark purplish red. The esophagus was patent and revealed no fistulous opening. The larynx was normally formed, but ended as a blind pouch at the lower border of the cricoid cartilage. The trachea was absent. The bronchi arose from a blind bifurcation from which point distally they were normally distributed (Text-Fig. A).

The heart was normal in size and outline, and the chambers were in normal relation. The interventricular septum was intact. The foramen ovale was anatomically patent. The pulmonary artery arose from the right ventricle, and after giving off the right and left pulmonary branches, continued as the large ductus arteriosus, which appeared to form the arch of the aorta. The aorta arose from the left ventricle, and gave off the usual three branches from the arch. Beyond the origin of the left subclavian artery the aorta continued as a very narrow vessel into the arch formed by the junction of the ductus arteriosus with the descending aorta. The ductus was at least twice the diameter of the isthmus (I/D = 0.5 or less) (Text-Fig. B).

The only other gross finding of note was the presence of both testicles in the pelvis.

Microscopic study of the lungs revealed marked edema of the interlobular septa. The bronchioles were distended and empty. The alveoli were more or less distended and contained red blood cells. The lining cells were inconspicuous. There was complete absence of aspirated

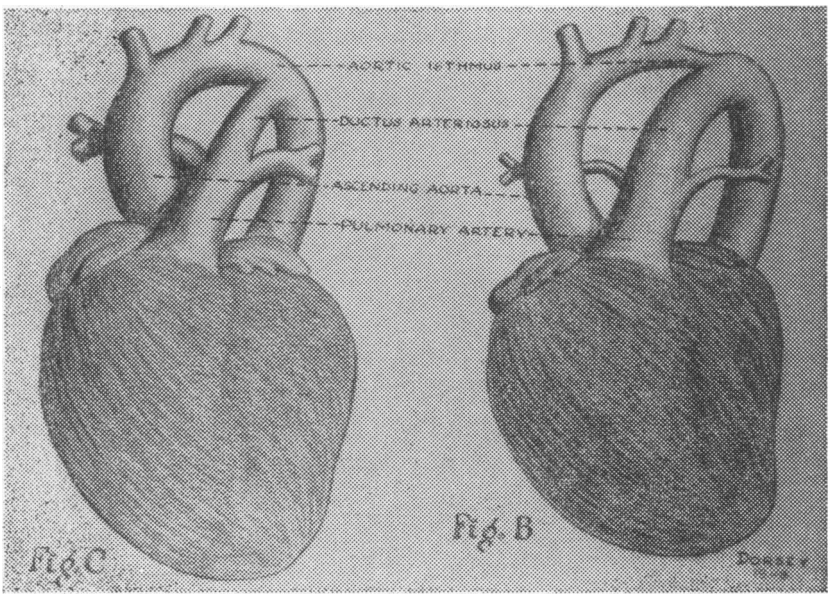
material. The alveoli were well formed, which bears out the observation of Potter and Bohlender[10] in their 2 cases of congenital malformations with no communication between the lung and upper air passages. The vessels were engorged. Microscopic study of transverse sections through the mediastinum at various levels failed to reveal any structure resembling tracheal tissue.



Text-Fig. A. Semi-diagrammatic representation of the tracheal defect in case 2.

Intra-uterine pulmonary expansion was obviously impossible, because there was no communication between the lungs and the larynx. The third portion of the aortic arch was less than half the diameter of the ductus arteriosus.

This case represents an extremely rare anomaly. Marek's case[13] had a fistula connecting the bronchial bifurcation with the esophagus. Fritz,[14] Walcher,[15] and others have reported cases in the foreign literature with fistulae between the esophagus and bronchial tree, combined with an absent, or partially absent, trachea. Payne[16] reported the only instance of absent trachea without communication to the esophagus to be found in the literature. Potter[11] studied a similar case. In neither instance were the great vessels measured.



Text-Figs. B and C. Semi-diagramatic comparison of normal isthmus/ductus size ratio (C) and abnormal ratio observed in cases 1 and 2 (B).

The anlage of the respiratory system appears in the human embryo of 23 segments (3.2 mm.) as a hollow, linear evagination of the ventral side of the primitive gut.[17] The more caudal portion appears first and forms an unpaired, pouch-like termination of the groove, the lung anlage. In normal development a constriction forms at the most caudal portion of the groove and progresses forward, the lateral borders of the groove fusing to form a septum between the tube of the primitive gut and the laryngotracheal tube with its terminal pouch.*

* Patten's[18] description of the development of the trachea and lungs suggests that the entire structure originates as an evagination of the caudal portion of the pharynx. However, the difference is probably one of terminology. The normally developed larynx in this case and in Payne's case[16] suggests that the proximal portion of the tract arises more or less independent of the trachea and the lung. Possibly in these cases separate evaginations appeared for the larynx and lung anlages with no connecting tracheal groove.

The more common anomalies of the trachea and esophagus include a fistula between the two passages with more or less complete failure of development of either the esophagus or trachea. They are explained on the basis of incomplete fusion of the margins of the primitive groove, and on differences in growth rates of the trachea and esophagus during subsequent development (Gruenwald[19]).

It seems logical to assume that the embryogenic defect in this case was a failure of the tracheal groove to form cephalad to the lung anlage, so that when constriction occurred the lung primordium was pinched off in the normal manner but left without communication to the pharynx.

Case 3

Case 3 was a female anencephalic monster, stillborn on the service of Dr. Swenson on June 6, 1947. The history of the pregnancy indicated that the infant was post-mature. Heart tones were last heard on May 7, 1947. Labor was induced, and delivery was spontaneous.

At autopsy, the body was that of a female anencephalic monster with complete spina bifida and a moderate omphalocele. The body weighed 950 gm. and measured 27 cm. from orbit to heel. There was beginning maceration.

The thoracic cavity was markedly diminished by the high position of the intact diaphragm. The thymus was broad, extending laterally into both pleural cavities, and increased in thickness, thus further diminishing the space available for the lungs. The lungs together weighed 3.3 gm. The left lung retained its early fetal configuration. The cut surfaces were firm and pink, speckled with gray. The trachea and main bronchi were patent and empty. The lung tissue did not float.

The heart weighed 5.6 gm. It was normally situated. The four cardiac chambers were normally formed. The interventricular septum was intact. The foramen ovale was anatomically patent. The great vessels arose normally. The ascending aorta was 8 mm. in diameter; the pulmonary artery and the descending aorta were each 7 mm. in diameter. The third portion of the aortic arch was 6 mm. in diameter and 2 mm. in length, and the ductus arteriosus was 5 mm. in diameter and 8 mm. in length.

The adrenals were markedly hypoplastic, and together weighed 0.5 gm.

The lungs appeared to be more hypoplastic than would be expected from the size of the body. The absence of the major portion of the head distorted both the weight and body length, so that these measurements did not serve as an accurate index of the stage of development.

To determine whether the lungs were more hypoplastic than the other organs, we computed the percentage of normal development that each of several organs had attained, with the results shown in Table I. It is evident that the lungs and adrenals were relatively less developed than the other organs. The failure of development of the lungs may be accounted for by the limitations of space in the pleural cavities as a result

TABLE I

Organ	Normal weight at term
	%
Thymus	39.3
Liver	39.0
Kidneys	37.7
Heart	27.3
Spleen	25.4
Adrenals	6.6
Lungs	5.7

of the high diaphragm and the transverse position of the thymus, or perhaps by primary failure of the lungs to develop, as is apparently true of the adrenals in anencephalic monsters.

In microscopic sections of the lungs the alveoli were small and lined by cuboidal epithelium in an abundant areolar stroma. In several areas, amorphous débris and pigment granules were visible in the bronchioles and air spaces. These could not be identified definitely as aspirated material.

In this case the third portion of the aortic arch was larger than the ductus arteriosus by a ratio of 6 to 5 (I/D = 1.2) in spite of the fact that the lungs were markedly hypoplastic.

COMMENT

The pulmonary arterial system of the fetal lung serves no apparent purpose other than supplying a portion of the nourishment of the lung. Therefore, provision is made for the major part of the circulating blood to by-pass this part of the vascular tree, shunts being provided by the foramen ovale and the ductus arteriosus.

Noback and Rehman[20] made a detailed dissection of the ductus arteriosus in fetal and newborn cadavers and stated that "The size of the ductus is much larger than is generally assumed, being equal to or greater in caliber than the aortic arch, the pulmonary artery, or the descending aorta." It is difficult to conceive of a branch (the ductus arteriosus) being larger than the trunk from which it originates (the pulmonary artery). In 26 fetal and newborn cadavers without gross abnormalities of the cardiovascular or respiratory systems, we obtained the measure-

ments shown in Table II, which do not agree with some of the findings of Noback and Rehman. Comparison of columns 3 and 4 shows that in only 6 of the 26 cases did the diameter of the ductus arteriosus exceed the diameter of the isthmus of the aortic arch, and in these cases the I/D ratio was 0.75 or greater.

In 11 instances the diameter of the aortic isthmus exceeded that of the ductus, and in the remaining 9 cases the two vessels were of equal size. The mean ductus size for the entire series was 5.8 mm. The mean isthmus size was 6.1 mm. The mean isthmus/ductus ratio for the series was 1.07 ± 0.27 (Text-Fig. C).

The I/D ratio in case 1 was 0.62, a deviation of 0.45 from the mean. The I/D ratio in case 2 was 0.50 or less, a deviation of 0.57 or more from the mean. While these differences are not great enough to be conclusive in so small a series, they are significantly suggestive. Anomalies preventing only pulmonary expansion are so rare as to preclude the accumulation of a large series in any one institution. Therefore we feel justified in publishing this report as a stimulus to other observers.

Barclay, Franklin, and Prichard[21] investigated the fetal and neonatal blood flow by means of kineradiographs of opaque media injected into the inferior and superior caval systems of fetal lambs delivered by cesarean section, injections being made both before and after interruption of the placental circulation and establishment of extra-uterine respiration. They demonstrated that all blood returning to the heart by way of the superior vena cava passes through the right side of the heart and out through the pulmonary artery to be distributed to the lungs and to the descending aorta through the ductus arteriosus. The stream of blood returning through the inferior vena cava is split by the anterior crescent of the foramen ovale, the greater part of the stream being diverted into the left atrium, and a lesser part continuing through the right side of the heart to be mixed with the blood from the superior vena cava. The blood entering the left heart from the foramen ovale and pulmonary veins is distributed to the coronary arteries, the head, and upper extremities by way of the branches from the aortic arch. What remains then joins the stream from the right heart at the junction of the ductus arteriosus with the aorta.

These studies indicate the paths of flow but do not give an accurate index of the relative amount of blood that passes through the right and left sides of the heart, a feature that is difficult to determine directly. Since the right ventricular muscle is approximately as large as the left at birth, we may infer that the intra-uterine work of the right heart has been equal to that of the left. On the other hand, in extra-uterine life

TABLE II

Actual and Proportionate Diameters of the Aortic Isthmus and Ductus Arteriosus in Fetuses and Newborn Infants

Case	Crown-heel length	Isthmus	Ductus	I/D ratio*	Cause of death	Anatomical diagnosis
	cm.	*mm.*	*mm.*			
1	16	2	2	1.00	Unknown	Macerated fetus; stillborn
2	21	3.5	2	1.75	Unknown	Stillborn
3	24	3	2	1.50	Intra-uterine sepsis	Septic placenta; interstitial pancreatitis; interstitial pneumonia; stillborn
4	27	4	3	1.33	Unknown	Macerated fetus; stillborn
5	28	4	3	1.33	Previability	Petechial hemorrhage about base of heart; prematurity
6	28	4	3	1.33	Meconium peritonitis	Meconium peritonitis; minimal interstitial pneumonia; stillborn
7	29	3	4	0.75	Unknown	Macerated fetus; stillborn
8	29.5	5	3	1.67	Previability	Prematurity; subepicardial petechiae; adrenal petechiae
9	30	4	3	1.33	Previability	Prematurity
10	33	8	8	1.00	Previability	Prematurity
11	40	6	4	1.50	Unknown	Petechial hemorrhages beneath visceral pleura; stillborn
12	42.5	7	5	1.40	Unknown	Minimal subtentorial hemorrhage; petechial hemorrhage, heart; pulmonary congestion
13	45	8	9	0.89	Intra-uterine sepsis	Petechial hemorrhage, heart and great vessels; pulmonary congestion; interstitial nephritis; acute myocarditis; stillborn
14	47	7	7	1.00	Unknown	Petechial hemorrhages of pleural cavities and heart, great vessels, and kidneys; stillborn
15	47	6	6	1.00	Intra-uterine asphyxia (premature separation of placenta)	Early acute glomerulonephritis; passive congestion of liver and spleen; pulmonary congestion; stillborn
16	47	7	7	1.00	Unknown	Petechial hemorrhage of pleura and pericardium; minimal intracranial hemorrhage; stillborn
17	49	9	10	0.90	Absence of kidneys and ureters	Absence of kidneys and ureters; pulmonary atelectasis and petechial hemorrhages; fibrosis of pancreas; interstitial myocarditis
18	49	5	5	1.00	Bilateral adrenal hemorrhage	Bilateral adrenal hemorrhage; petechial hemorrhage in heart, lungs, and kidneys; stillborn
19	49	8	6	1.33	Intracranial hemorrhage	Intracranial hemorrhage; cleft palate and harelip; subpericardial and subpleural petechial hemorrhage; hemorrhage into thymus; stillborn
20	49	8	8	1.00	Unknown	Petechial hemorrhages about the heart and great vessels; stillborn
21	51	8	9	0.89	Intracranial hemorrhage	Intracranial hemorrhage
22	51.5	8	8	1.00	Unknown	Petechial hemorrhage beneath serosa of heart, lungs, and about the thymus; stillborn
23	52	8	7	1.14	Meconium peritonitis	Meconium peritonitis; early bronchopneumonia; macerated fetus; stillborn
24	52	7	9	0.78	Intracranial hemorrhage	Intracranial hemorrhage; petechial hemorrhage, pericardium; polydactylism; Meckel's diverticulum; bronchopneumonia; stillborn
25	52	8	9	0.89	Unknown	Subpleural and subpericardial petechial hemorrhage; macerated fetus; stillborn
26	53	8	8	1.00	Neonatal sepsis	Bronchopneumonia; interstitial nephritis; interstitial myocarditis

* Diameter of the isthmus: diameter of the ductus arteriosus.

the muscle mass of the left ventricle rapidly exceeds that of the right ventricle, even though they pump equal volumes of blood, so we may assume that the postnatal work demand upon the right heart is correspondingly less than that placed on the left.

The usual explanations of fetal circulation are predicated upon a functionless fetal lung. The small demand for blood of a functionless lung should require only a fraction of the output of the fetal right heart, and therefore we would assume that both ventricles contribute about equally to the systemic and placental circulation. Since a relatively large proportion of the blood from the left ventricle is supplied to the head and upper extremities, while the stream from the right ventricle supplies only the lungs before contributing blood to the descending aorta, we would expect that, if only sufficient blood passed through the pulmonary vascular tree to nourish the lung, relatively more of the blood in the descending aorta would come from the right heart than from the left.

Fetal blood vessels vary in size according to the average intravascular pressure and volume of blood flow as evidenced by the alterations in relative vessel size in certain congenital lesions of the heart. For instance, in aortic valve atresia there is hypoplasia of the aorta. Thus vessel size is an indicator of the average relative pressure and volume flow carried by the vessel. On this basis, if the lung were normally inactive in utero, we would expect the ductus arteriosus to be larger than the isthmus of the aorta, since the volume of blood flowing through the ductus would be greater. This is not the usual finding. Abel and Windle[22] studied the blood content of fetal lungs before the beginning of respiration and after 1 to 24 hours of air breathing by calculating the hemoglobin content from determinations of the iron content of the ashed lung. They found no appreciable difference in the amount of blood in the lung before and after air breathing. They concluded that there is no sudden increase in the volume of pulmonary circulation with air breathing. They stated that "It appears that a circulation is already present in the lungs during late prenatal life which is wholly capable of caring for oxygenation pending the assumption of respiration." Therefore, we must look for other anatomical and physiologic factors in the later stages of development to explain the increasing diversion of blood into the pulmonary circulation.

Since, in general, organ systems that must function at birth show evidence of trials of function in utero (urine in the bladder, bile in the gallbladder, swallowed amniotic contents in the stomach), and organs that cannot function in utero (the eyes, for instance) show slow devel-

opment of function post-natally, it is reasonable to expect that an organ system so important to the survival of the individual as the respiratory apparatus should undertake intra-uterine trials of function. The portions of the respiratory apparatus that would be expected to benefit by intra-uterine respiratory movements are the chest muscles, the pulmonary alveoli, and the pulmonary vascular system. The effect upon the muscles would be difficult to evaluate. Potter and Bohlender[10] have shown that alveolar development does not depend upon the expansion of the lung, and our findings agree with theirs. We are concerned, then, with pulmonary vascular development.

After birth the change in intrapulmonary pressure with the onset of regular, rhythmic respiration is assumed to play a major rôle in diverting the total output of the right heart into the pulmonary circulation. The studies of Barclay, Franklin, and Prichard[21] show that the first change to take place in blood flow after birth is cessation of flow through the foramen ovale. This occurs almost immediately with the beginning of extra-uterine respiration. Flow through the ductus arteriosus continues for a few minutes longer, and stops more gradually.

The closure of the foramen ovale is accomplished by an increased flow of blood from the pulmonary veins, which increases the pressure in the left atrium and forces the posterior leaf of the foramen ovale against the anterior crescent to occlude the opening. Since this occurs before the flow in the ductus ceases, the expansion of the lungs and the resultant changes in pressure must play the major rôle in diverting blood from the ductus arteriosus into the pulmonary system. If pressure against the walls of the ductus, as suggested by Noback and Rehman,[20] or active contraction of the ductus, as suggested by Patten[18] and others, were the primary factors, we would expect the flow in the ductus to stop first.

Since deep, rhythmic, extra-uterine respiration produces profound changes in the circulatory system, it is reasonable to expect that intra-uterine respiratory movements, even though less active, would also have an effect on the pulmonary circulation of the fetus. Acting over a much longer period of time, this should produce changes in the relative blood flow through, and size of, the great vessels. Absence of intra-uterine respiratory movements should negate this effect.

It is hardly conceivable that a vascular network capable of handling the entire circulating volume of blood immediately after birth could develop without stimulus. Abel and Windle[22] pointed out that "studies by Thoma in 1894 (Keibel and Mall, '12) and later by Mall ('06) and

Flint ('06) have shown that arteries and veins are formed from capillary nets and represent a functional adaptation of the net to the demands of the circulation. In other words it is developmentally unsound to assume the existence of well developed pulmonary vessels which contain a volume of blood well below their capacity. The size of these vessels depends entirely on the volume of blood coursing through them and is a resultant adaptation thereto."

Hamilton, Woodbury, and Woods[23] stated: "To conceive of a capacious but virtually unused vascular plexus developing in the growing lungs—ready at the instant of birth to receive a suddenly re-routed current of blood carried to the lungs for oxygenation"[24]—may well smack of teleology. It seems more logical, therefore, to expect the development of the pulmonary vascular system to take place in response to gradually increasing functional demand. This demand could consist, most logically, of intra-uterine respiration-like movements resulting in the expansion of the lungs.

With the stimulus of lung expansion we would expect an increasing diversion of blood into the pulmonary vascular system with less blood remaining in the ductus arteriosus. This would balance the withdrawal of blood through the branches of the aortic arch and equalize the volume flow through the aortic isthmus and the ductus arteriosus. This would explain the usual size ratio between these two vessels.

If this is the case, interference with expansion of the lungs during intra-uterine life should result in an underdeveloped pulmonary vascular tree, which in turn should be reflected in a relative increase in the size of the ductus arteriosus, since less blood would circulate in the lungs and more would pass through the ductus. In case 1, one lung was hypoplastic and the infant was presumably unable to expand the lungs in utero because the absent diaphragm made it impossible to reduce the pressure within the chest. In this case the diameter of the ductus arteriosus exceeded the diameter of the aortic isthmus by a ratio of 8 to 5 (I/D ratio, 0.62) (Text-Fig. B, see page 415). It might be argued that the hypoplasia of the lung was entirely responsible for the reduced pulmonary blood flow.

In case 2, the lungs were of normal size and the alveoli were normally developed. The only functional impairment was interference with expansion of the lungs, and in this case the aortic isthmus was less than half the diameter of the ductus arteriosus (I/D ratio, 0.5 or less) (Text-Fig. B, see page 415). This suggests a markedly diminished pulmonary vascular development. Since the only function rendered impossible as

a result of the absent trachea was expansion of the lungs, we infer that the diminished pulmonary vascular development was the result of the absence of lung expansion. The fact that the aortic isthmus was reduced in size absolutely, as well as relatively, in both cases, suggests that a considerable proportion of the output of the left heart is derived from the return flow through the pulmonary veins, and if this is diminished for any reason, the flow through the foramen ovale is insufficient to maintain a normal volume flow through the left heart.

In case 3, the normal size ratio between the ductus arteriosus and the aortic isthmus was maintained in spite of hypoplastic and supposedly functionless lungs. However, in this case the brain, with its relatively enormous demand for blood, was absent. Less blood was withdrawn from the aorta and thus a larger blood volume was permitted to flow through the aortic isthmus, balancing the increased flow through the ductus arteriosus.

Work now in progress on certain arteriosclerotic changes in the ductus arteriosus appearing in the later months of gestation seems to support our theory further and will be reported in a later paper.

Patten[18] stated that "If, through any deficiency in the development of the lungs themselves, or of their vessels, undue resistance to the free passage of blood exists, the pulmonary artery will continue to shunt blood over the ductus arteriosus to the aorta just as it did before birth. That this happens but rarely is perhaps the most remarkable fact about the entire series of circulatory changes which take place at birth; for there is no way this peripheral part of the pulmonary circulation can be tested out under functional conditions while the fetus remains in utero."

We believe that intra-uterine respiration-like movements resulting in expansion of the lungs and the aspiration of amniotic fluid not only constitute a test of the peripheral pulmonary circulation under functional conditions in utero, but are necessary for the development of the pulmonary vascular system, making possible an adequate post-natal pulmonary circulation.

Summary

We have presented a hypothesis of a "functioning" fetal lung as a necessary stimulus to the development of the pulmonary vascular system. This hypothesis is based upon analysis of facts already known and upon measurements of the great vessels in 26 cases of fetal and neonatal death without pulmonary or cardiac anomalies. It is supported by a case with hypoplasia of one lung and interference with respiration due

to absence of the left diaphragm. There was an altered size ratio between the aortic isthmus and the ductus arteriosus, I/D = 0.62. The hypothesis is further supported by a second case with grossly normal pulmonary development but with absence of the trachea, which made expansion of the lungs impossible. Again the I/D was altered, *i.e.*, 0.5.

A third case, an anencephalic monster with an anomaly of the respiratory system that would not prevent intra-uterine respiration-like movements, showed normal vascular development. The absence of respiratory movements reported[9] with this condition is assumed to have been counterbalanced by the decreased circulation to the head.

Thus supported, our hypothesis becomes a theory. The theory will require many observations to confirm it; but on the basis of this study we offer the following conclusions subject to further verification.

Conclusions

The normal isthmus/ductus diameter ratio is approximately 1.

The average volume flow through the pulmonary vascular system prenatally is approximately equal to the volume of blood supplied to the head and upper extremities.

This suggests an active stimulus to pulmonary blood flow, which could consist, most logically, of respiration-like movements resulting in lung expansion with its concomitant reduction in pressure in the pulmonary vascular system.

The vascular anomaly in the first 2 cases indicates an alteration in blood flow resulting from a decreased pulmonary circulation. Since it was found to be associated with a malformation that prevented only intra-uterine pulmonary expansion, we conclude that interference with intra-uterine pulmonary expansion can be the sole cause of this vascular anomaly. If interference with a function produces abnormal development, that function must be physiologic. Therefore, intra-uterine respiration-like movements are a physiologic function, because free expansion of the lungs during fetal life is necessary for normal development of the pulmonary vascular system.

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